

Spindle and Bearing Maintenance Guide

Synthetic project reference document for Phase 5 RAG testing.

1. Spindle Bearing Function

Spindle bearings support high-speed rotation and maintain shaft alignment. Bearing condition strongly affects vibration, temperature, tool runout, and surface finish.

2. Symptoms of Bearing Degradation

Common indicators include increasing vibration, rising bearing temperature, unusual mechanical noise, increased motor current, unstable spindle behavior, and worsening machining quality. A combination of increasing vibration and temperature is more significant than either measurement alone.

3. Possible Causes

Possible causes include insufficient lubrication, contaminated lubricant, bearing wear, incorrect preload, shaft misalignment, excessive load, tool imbalance, and incorrect installation. Sensor trends should be evaluated over time.

4. Inspection Procedure

Follow the site's energy-isolation procedure before physical inspection. Check lubricant condition, bearing housing temperature, mechanical looseness, shaft condition, tool runout, and coupling or drive alignment.

5. Corrective Actions

Corrective action may include lubrication service, cleaning, alignment correction, balancing, tightening, or bearing replacement depending on the confirmed cause. Do not replace a bearing solely from one sensor value.

Important: This is synthetic reference content created for software testing and demonstration. It is not a manufacturer manual or a substitute for approved maintenance procedures, qualified personnel, or applicable safety requirements.